

LPSRGAN: Generative adversarial networks for super-resolution of license plate image

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ABSTRACT

This paper proposes a super-resolution algorithm for reconstructing license plate images based on generative adversarial networks (GAN) for improving the recognition rate of low-resolution license plate images. To this end, this paper first designs a new image degradation model, called the n-stage random combination degradation model (n-RCD), which extends the traditional degradation model horizontally and vertically to better simulate the features of low-resolution license plate images in natural scenes. Subsequently, to address the problem that existing GAN models cannot adapt to the complex degradation space in natural scenes, this paper proposes a new network structure called LPSRGAN, which optimizes the SRGAN model from the generator and discriminator to improve its super-resolution capability. In addition, this paper also proposes a perceptual optical character recognition (OCR) loss for license plate images, which defines the connectionist temporal classification (CTC) loss of the output of the OCR network with character label values to better preserve the character details and features of the license plate images. Finally, the experiments reported in this paper show that the proposed algorithm improves the recognition rate of the original low-resolution license plate image by 12.48%, and the recognition rate of the reconstructed image reaches 93.90%. Based on this, this paper also tests the algorithm on actual captured data, and the results show that it performs well in natural scenes, demonstrating the algorithm has broad application prospects.

1. Introduction

As the most important information carrier of a car, the license plate is crucial in intelligent transportation and criminal investigation tasks. However, due to the limitation of acquisition equipment and the interference from complex environments, license plate images often suffer from blurring and distortion, seriously affecting the accuracy and efficiency of tasks such as license plate detection and license plate recognition. Therefore, license plate image super-resolution (SR) has become an issue of great concern in intelligent transportation systems.

Recently, the emergence of generative adversarial networks (GANs) [8] has provided an effective solution to the image SR problem. Adversarial learning enables GANs to generate high-quality images, avoiding the traditional manual feature extraction and manual design process, making the image SR task more efficient and effective. Various GAN-based Single Image Super Resolution (SISR) methods [1,2,9–14] have been proposed to improve the visual quality of SR results. However, there is still a significant difference between the SR results of these methods and authentic images because they all use ideal bi-trivial downsampling [15] or conventional degradation [16] as degradation

models, which are inadequate to represent the actual complex degradation space. For this reason, some works [17–22] trained models by adjusting the camera focal length to capture low resolution (LR) and high resolution (HR) images separately for image alignment to form the actual degraded dataset. However, this method suffers from two problems. First, the alignment error cannot be eliminated, resulting in the inability to create a good LR-HR image pair. Second, the image-capture equipment limits this approach, and if the equipment does not match the dataset acquisition device, it will lead to a drastic degradation of performance. Other works [3,4,23,24] have achieved the same goal by designing more realistic degradation models. Although these works have achieved good results on some public datasets and cartoon images, they have not achieved satisfactory results on actual degraded complex license plate images.

In this paper, in order to solve the SR problem of license plate image, we first extend the traditional degradation model for horizontal and vertical directions, and design an n-stage random combination degradation (n-RCD) model, which better models the LR, blurring, and distortion of license plate images. We then propose a new GAN

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(a) LR		沪H273HQ
(b) ESRGAN		粤H27380
(c) ESRGAN+		粤H2738Q
(d) BSRGAN		桂H273BA
(e) Real-ESRGAN		桂H273BQ
(f) TSRN		皖HZ1388
(g) RTSRN		皖H21F88
(h) MARCONet		粤H275HH
(i) LPSRGAN(ours)		甘H273BQ
(j) HR		甘H273BQ

Fig. 1. Comparison of SR results generated from ESRGAN [1], ESRGAN+ [2], BSRGAN [3], Real-ESRGAN [4], TSRN [5], RTSRN [6], MARCONet [7], LPSRGAN(ours). The right column shows the license plate recognition results.

network, called LPSRGAN, to adapt to the more complex degradation space of the degradation model. Compared with the previous GAN networks, LPSRGAN introduces additional jump connection and residue dense block (RDB) feature aggregation modules in the generator and Dropout layer before the last convolutional layer. These introductions help to improve the capacity, feature representation capability, and generalization performance of LPSRGAN. In addition, this paper proposes a perceptual optical character recognition (OCR) loss for license plate images, which enables the generator to produce license plate images with more apparent characters by jointly optimizing the generator and discriminator. Fig. 1 shows the performance of our model.

In summary, the contributions of this paper are as follows:

- An n-RCD is designed, extending the traditional degradation model for horizontal and vertical directions, and better simulates the actual degradation process of license plate images.
- A new GAN network, LPSRGAN, is proposed to improve the network's capacity, feature representation capability, and generalization performance, enhancing the visual performance of its reconstructed images.
- A downstream task-based perceptual OCR loss is introduced for license plate images, which improves the character clarity of reconstructed license plate images.

2. Related work

Since the focus of this paper is to design a GAN model to reconstruct the actual degraded license plate image, we briefly overview the relevant degradation models, single image SR, and license plate image SR methods.

2.1. Degradation model

In earlier SR work [25–27], a simple dual-triple downsampling [15] method was typically used for the acquisition of training and test data. Slightly more complex work [27,28] used classical degradation models [16] consisting of blur, downsampling and noise, which are inadequate for the simulation of complex real-world degraded images. Recently,

Zhang et al. [3] devised a more complex but closer to realistic degradation model, including random permutations of blurring, downsampling, and noise degradation, as well as introducing a more realistic camera sensor and JPEG compression noise during noise addition. However, Wang et al. [4] considered the role of random permutation doubtful. Instead, they designed a higher-order degradation model by several iterative degradation processes and considering familiar ringing and overshoot artifacts, capable of covering a larger degradation space. Sahak et al. [29] further abstracted the noise from the degradation process and applied noise modulation enhancement to enhance the model denoising effect.

In conclusion, current available image degradation models can provide good simulations of simple degradation. However, there is still a mismatch between the assumed and accurate image degradation models for complex degradation in natural scenes. Thus, it is essential to develop a realistic degradation model that can be used to train the super-resolution model for real-world applications.

2.2. Single image super-resolution

Convolution Neural Network (CNN)-based methods have made significant progress in reconstructing HR images in recent years. Dong et al. [25] proposed the SRCNN network which achieved better performance than all conventional methods using only a simple three-layer CNN architecture. Since then, many CNN-based SR models have been proposed. These models can be divided into two categories, i.e., signal fidelity-oriented models [30–33] and perceptual quality-oriented models [1–4,11].

Signal fidelity-oriented models focus on minimizing the mean square reconstruction error (MSE) as a means to optimize the peak signal-to-noise ratio (PSNR) and structural similarity (SSIM) of the reconstructed images. Since SRCNN [25], more network architectures have been used in the field of SISR, such as residual connectivity networks [30], dense connectivity networks [34], Laplace pyramid networks [35], recursive networks [31] and residual dense networks [36]. The reconstructed images obtained by such models have high PSNR, but they usually lack high-frequency details and are perceptually unsatisfactory.

Perceptual quality-oriented models mainly focus on improving the high-frequency details and perceptual quality of reconstructed images, usually with GANs as the basic architecture. Ledig et al. [9] proposed SRGAN with adversarial training on top of SRResNet generators, which can recover realistic image textures. They became the pioneering work of GAN in the field of SISR. To further improve the visual quality, Wang et al. [1] enhanced the network structure, adversarial loss, and perceptual loss in SRGAN. They proposed ESRGAN, in which the generator architecture RRDB was widely used in other GAN-based SR models [3,4,11]. Wang et al. [4] further improved the discriminator of ESRGAN and trained the model using highly simulated synthetic data to propose Real-ESRGAN, which achieved the best visual quality. In addition, Sahak et al. [29] proposed a diffusion model SR3+ for blind SR, which obtained state-of-the-art FID scores on a blind SR test dataset.

Although existing SISR models are sufficient for most images in publicly available datasets, to our knowledge, no SISR model can be easily applied to natural image SR in general. Therefore, designing a SISR network with high generalization performance and further mining the mapping relationship between LR images and HR images is essential for the current SR work from theory to practice.

2.3. License plate image super-resolution

Several works exist focusing on the SR of text images [37–41]. As a type of text image, the SR of license plate images has been less studied. Liu et al. [42] proposed a license plate super-resolution framework that incorporates progressive vehicle search and domain prior GAN. This framework leverages information from multiple LR images to enhance

the recovery of license plate numbers. Inspired by GoogLeNet, Yang et al. [43] designed a SISR network based on the Inception architecture for license plate image recovery. In order to tackle the issue of excessively smooth characters in the reconstructed images, Lai et al. [44] employed SRGAN for license plate images, resulting in a notable enhancement in character clarity. In addition, Lee et al. [45] proposed a character-based SRGAN with better character reconstruction than the original model. Considering the noise problem of license plate images, Hamdi et al. [46] proposed a dual generative adversarial network with image denoising and image SR cascading for image reconstruction, effectively improving the license plate recognition rate. Recently, Hijji et al. [47] extended SRGAN by adding an intelligent motion blur method that helps to improve image resolution and remove motion blur from a given image.

In summary, the current work on the SR of license plate images is still at the stage of simple application of the model, without improvement for the characteristics of license plate images, and is thus ineffective in practical application. Therefore, it is necessary to modify the degradation model, network structure, loss function, and other contents for license plate images.

3. Proposed method

3.1. n -stage random combination degradation model

The goal of blind SR is to reconstruct HR images from LR images with no-known degradation, usually using a conventional degradation model [15] consisting of blur, downsampling, and noise to synthesize LR images, which can be mathematically modelled by

$$I_{LR} = (I_{HR} \otimes k) \downarrow_s + n. \quad (1)$$

It is presumed that the LR image I_{LR} is produced by first making a convolution with a fuzzy kernel k , then a downsampling operation with a scale factor s , and finally adding noise n .

Traditional degradation models achieve the expected results when faced with simply degraded samples. However, their performance dramatically drops when applied to images with actual complex degradation. This significant drop in performance occurs because the actual degradation encompasses a much larger degradation space than the traditional degradation model. Thus, this paper uses a degradation model with a larger degradation space to simulate more realistic degradation.

Before presenting our degradation model, we review the traditional degradation model and provide the rationale of our model. First, the traditional degradation model only incorporates the three basic degradations due to blur, downsampling and noise, limiting the degradation space. Thus, this paper considers expanding the degradation space horizontally and vertically. Noting that real degradation process is much more diverse, including but not limited to camera sensor degradation, network transmission degradation, JPEG compression, etc., and inspired by BSRGAN [3] and Real-ESRGAN [4] we extended the degradation space horizontally by adding degradation due to JPEG compression; camera sensor, and ringing effect.

Second, the three basic degradations of the traditional degradation model are only performed once, which we call one-stage degradation. However, a real image may undergo multiple degradation processes, e.g., transmission from one end device to another end device through numerous data links. Furthermore, unlike traditional degradation models that apply degradation factors in a fixed order, we randomly combine these degradation factors, thus changing the nature of the degradation. Degradation situations that rarely occur in real scenes help to further expand the degradation space and improve the generalization ability of the model. Thus, vertically we extend the one-stage degradation model to an n -stage degradation model, i.e., the degradation model contains several repeated random degradation processes to simulate the complex combination of different degradations

that occur in real LR images. It is not difficult to find that other degradation models (e.g., traditional degradation models, and the degradation models in BSRGAN [3] and Real-ESRGAN [4]) are only subsets of n -RCD.

The proposed degradation process of one stage can be represented by

$$I_{LR} = D(I_{HR}) = I_{HR} \odot \{B \odot D \odot N \odot J \odot S \odot R\}_{Random}. \quad (2)$$

The LR image, I_{LR} , of the degradation model in a single stage is formed by the HR image I_{HR} after a random combination of blurring B , downsampling D , noise N , JPEG compression J , camera sensor degradation S , and ringing effect R .

Vertically, we extend the one-stage degradation model into an n -stage degradation model. The n -stage degradation model contains n one-stage degradation processes, i.e.,

$$I_{LR} = D_1 \odot D_2 \odot \dots \odot D_n(I_{HR}), \quad (3)$$

where $D_i (i = 1, 2, \dots, n)$ denotes the degradation process of the i th stage. As a result, we obtain the n -RCD. Fig. 2 shows the overall architecture of the degradation model.

3.1.1. Blur

In the traditional degradation model, blurring is achieved by convolving the HR image with a blur kernel, where the blur kernel is usually chosen to be Gaussian blur, i.e.,

$$F(x, y) = \frac{1}{K} \exp\left(-\frac{1}{2} C^T \Sigma^{-1} C\right), \quad C = [x, y]^T, \quad (4)$$

where C denotes the spatial coordinates, K is a normalization constant, and Σ is the covariance matrix, i.e.,

$$\Sigma = \begin{bmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{bmatrix} \begin{bmatrix} \sigma_x^2 & 0 \\ 0 & \sigma_y^2 \end{bmatrix} \begin{bmatrix} \cos \theta & \sin \theta \\ -\sin \theta & \cos \theta \end{bmatrix}, \quad (5)$$

where σ_x and σ_y respectively represent the standard deviation of the x -axis and y -axis, θ represents the rotation angle, $\sigma_x = \sigma_y$ represents an isotropic kernel, and otherwise it is an anisotropic kernel.

Although the point spread functions of many optical measurement systems and imaging systems, tend to be Gaussian blur functions, the Gaussian blur does not fully simulate actual camera blur. Based on the experience of Wang et al. [4], we further add a generalized Gaussian blur kernel and a plateau-shaped blur kernel to the degradation model, thus increasing the blur degradation space. Their probability density functions are $\frac{1}{K} \exp\left(-\frac{1}{2} (C^T \Sigma^{-1} C)^\beta\right)$ and $\frac{1}{K} \frac{1}{1+(C^T \Sigma^{-1} C)^\beta}$, respectively,

where β is the shape parameter. Fig. 3 shows various blur kernels.

3.1.2. Downsampling

Downsampling is one of the three basic degradations of the traditional degradation model, and the commonly used algorithms include nearest neighbour interpolation, area-based interpolation, bilinear interpolation and bicubic interpolation, each producing different effects. Since we need to degrade the image in n stages to keep the image resolution within a reasonable range, each stage performs random upsampling and downsampling operations on the image, and the interpolation algorithm is chosen randomly among area-based interpolation, bilinear interpolation, and bicubic interpolation to avoid pixel misalignment from nearest-neighbour interpolation [48].

Area-based interpolation is a simple interpolation method that regionally samples the pixels of the source image and then assigns the average of these pixels to the pixels of the destination image. It is computed by

$$I_{dst}(x, y) = \frac{1}{4} [I_{src}(x_1, y_1) + I_{src}(x_1, y_2) + I_{src}(x_2, y_1) + I_{src}(x_2, y_2)], \quad (6)$$

where (x_1, y_1) , (x_1, y_2) , (x_2, y_1) and (x_2, y_2) are the four closest pixel coordinates to (x, y) in the source image, I_{src} is the pixel value in the source image, I_{dst} is the pixel value in the destination image.

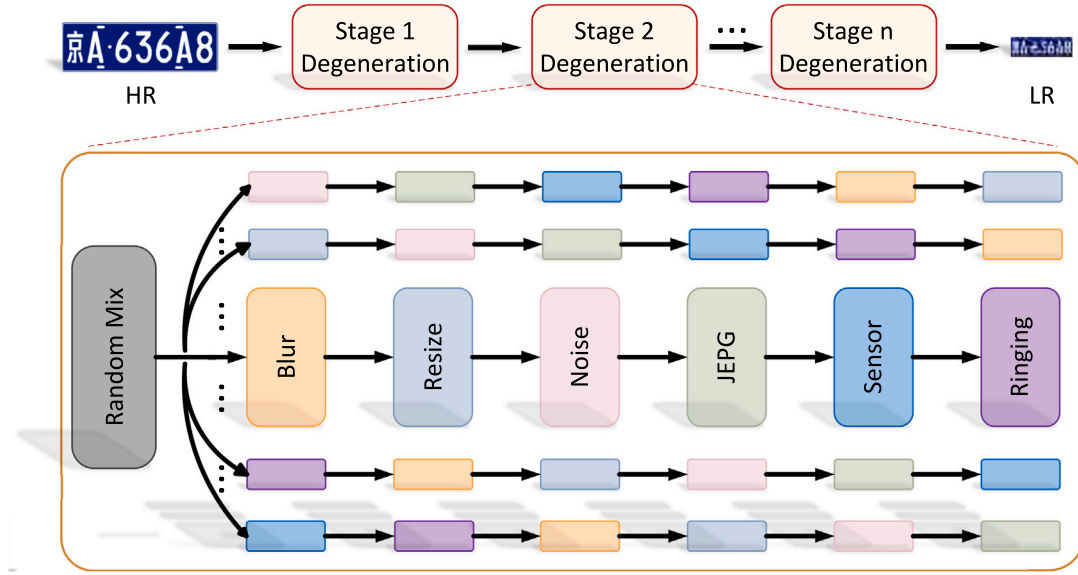


Fig. 2. Overall architecture of the n-stage stochastic combination degradation model.

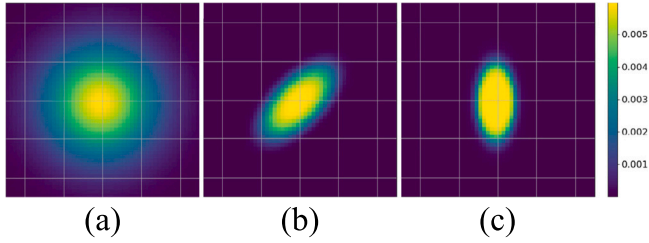


Fig. 3. Blur kernels: (a) Gaussian blur kernel; (b) Generalized Gaussian blur kernel; and (c) Plateau-shaped blur kernel.

Bilinear interpolation calculates the value of a target pixel based on the values of the four nearest neighbour pixels on the image and the offset of the target pixel on the source image coordinates. It is computed by

$$I_{dst}(x, y) = (1 - \alpha)(1 - \beta)I_{src}(x_1, y_1) + \alpha(1 - \beta)I_{src}(x_2, y_1) + (1 - \alpha)\beta I_{src}(x_1, y_2) + \alpha\beta I_{src}(x_2, y_2), \quad (7)$$

where $\alpha = x - x_1$ and $\beta = y - y_1$ are the offsets of (x, y) relative to (x_1, y_1) and (x_1, y_2) .

Bicubic interpolation is an advanced interpolation method that uses a 4×4 pixel block (usually 16 pixels) to estimate the value of the target pixel, and then interpolates through a cubic polynomial. It is computed by

$$I_{dst}(x, y) = \sum_{i=0}^3 \sum_{j=0}^3 H(i) \cdot H(j) \cdot I_{src}(x_i, y_j), \quad (8)$$

where (x_i, y_j) are the pixel coordinates in the 4×4 pixel block in the source image, $H(i)$ and $H(j)$ are the weighting functions of bicubic interpolation, usually a cubic polynomial. The specific form of these weight functions can vary according to different bicubic interpolation algorithms.

3.1.3. Noise

When there is no information about the noise, the Gaussian noise assumption is the most conservative choice. Gaussian noise follows

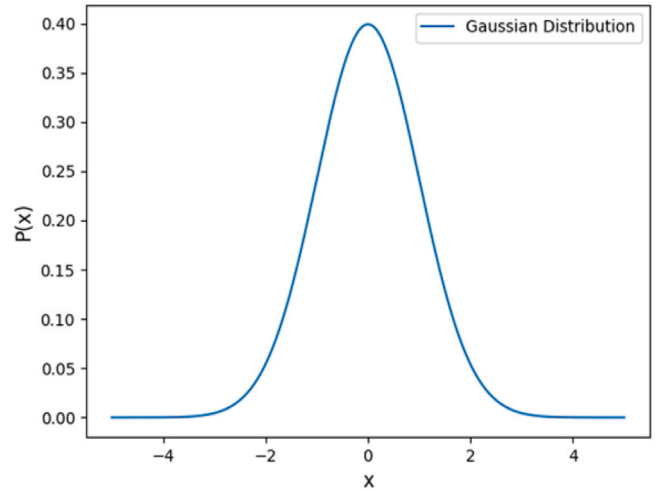


Fig. 4. Gaussian distribution ($\mu = 0, \sigma = 1$).

a Gaussian distribution, and the level of noise is determined by its standard deviation, as shown in Fig. 4. We use two types of additive Gaussian noise in the degradation model to simulate the noise generation process. The first type is colour noise, which adds independent sampling noise to the three channels of an RGB image. The second type is grey noise, which adds the same sampling noise to the three channels of the RGB image. The sampling noise of a single channel follows a Gaussian distribution, i.e.,

$$P(x) = \frac{1}{\sqrt{2\pi}\sigma^2} \exp\left(-\frac{(x - \mu)^2}{2\sigma^2}\right), \quad (9)$$

where x represents the grey value, μ and σ represent the mean and standard deviation, respectively.

3.1.4. JPEG compression

JPEG compression is a commonly employed method for reducing image file sizes, aiming to conserve bandwidth. However, it often introduces block artifacts and noise, particularly when high compression ratios are used. The level of compression is controlled by a quality factor q , which ranges from 0 to 100. A lower value of q indicates a



Fig. 5. JPEG compressed license plate image ($q = 10$).

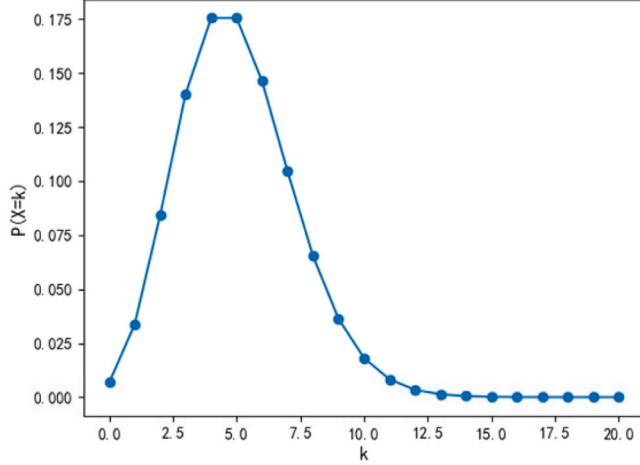


Fig. 6. Poisson distribution ($\lambda = 5$).

higher compression ratio and, consequently, lower image quality. We simulate the process of JPEG compression on various channels within the YCbCr colour space. Fig. 5 shows an image after JPEG compression, in which obvious blocking artifacts and noise can be seen.

3.1.5. Camera sensor degradation

Non-Gaussian noise is prevalent in image signal processing (ISP) pipelines for modern digital cameras. For this reason, Zhang et al. [3] used several digital cameras and cell phones to build an ISP pipeline to simulate non-Gaussian noise in degraded images. However, this approach is challenging, and the equipment limits the degradation space.

Inspired by Wang et al. [4], we use Poisson noise to model the degradation of camera sensors. Poisson noise follows a Poisson distribution, as shown in Fig. 6, where the level of noise is proportional to the brightness of each pixel in the image, i.e.,

$$P(X = k) = \frac{1}{k!} \exp(-\lambda) \lambda^k, \quad (10)$$

where λ determines the intensity of the noise, and its value is defined as the number of unique pixel values in the image. This is to adaptively generate Poisson noise according to the characteristics of the image. k is a non-negative integer representing the grey value of a certain pixel.

Additionally, the noise at different pixels is independent of each other. This type of noise is often employed to represent sensor noise arising from statistical quantum fluctuations.

3.1.6. Ringing effect

The ringing effect frequently occurs as undesired artifacts in images, presenting as false edges near abrupt changes or sharp transitions. This phenomenon is widespread and can be observed in various images. Inspired by Wang et al. [4], we use sinc filters in each stage of degradation to model the ringing effect. Sinc filters are filters constructed from sinc functions, and the normalized sinc function is

$$\text{Sinc}(x) = \frac{\sin(\pi x)}{\pi x}. \quad (11)$$

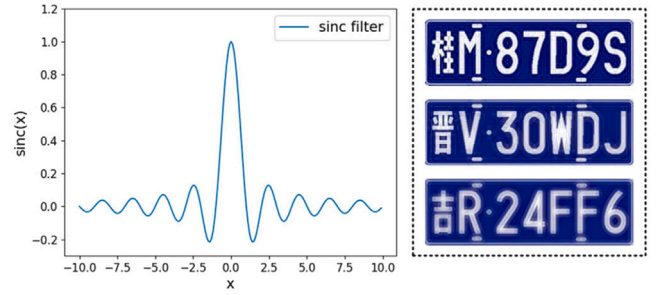


Fig. 7. Sinc filter and the resulting ringing effect in example images, where the effect is more pronounced from the top image to the bottom image, at the edges of characters and the zoomed parts of an image.

Fig. 7 shows the sinc filter and its synthesized ringing effect in license plate images.

3.2. Network structure

In this paper, we propose a new SISR model called LPSRGAN, using GAN as the basic architecture of the model. The model comprises a generator and discriminator, and requires a strategy of model training.

3.2.1. Generator

To accommodate a larger degradation space and further improve the recovered image quality of the GAN-based model, we made the following two modifications to the structure of the generator, as shown in Fig. 8. First, we replaced the commonly used Residual in Residual Dense Block (RRDB) with the improved Residual in Residual Dense Block (RRDB+) proposed in this paper. RRDB+ replaces the dense block (DB) in RRDB with the RDB, and introduces the RDB feature aggregation module. Second, we added a dropout layer before the last convolutional layer of the generator. The improved generator is called RRDBNet+.

We retain the overall architecture of ESRGAN and replace its basic blocks from RRDB with the proposed RRDB+. Specifically, we added residual connections from input to output to the dense block (DB) of RRDB to increase the network capacity without increasing its complexity, and the resulting new module is called RDB. In addition, we introduce RDB feature aggregation module in RRDB by adding the output features of each RDB directly to the end of the RRDB+ module via the weight parameter β . Compared with the approach of simply stacking multiple RDB modules, our method fully utilizes the output features of each RDB without any loss, resulting in a more discriminative feature representation.

Kong et al. [49] have shown that the appropriate use of dropout layers benefits SR networks and improves the generalization ability. Thus, we added a channel-wise dropout before the last convolutional layer of the generator to enhance the generalization ability of the network while improving the channel representation capability and significantly improve the stability of the model under complex degradation.

3.2.2. Discriminator

Due to the larger degradation space, the discriminator of Real-ESRGAN needs considerably more discriminative power to handle the complex training output. As a result, Wang et al. [4] improved the VGG discriminator in ESRGAN to a U-Net with skip connections. We further adopted a U-Net3+ network [50] with full-scale jump connections as a discriminator, as shown in Fig. 9.

The U-Net3+ network structure is divided into three parts: encoder (Down-sampling), decoder (Up-sampling), and Full-Scale Skip Connection. The encoder (Down-sampling) and decoder (Up-sampling) are respectively the same as down-sampling and up-sampling of the U-Net network. The construction of the feature map, X_{De}^3 , is similar to

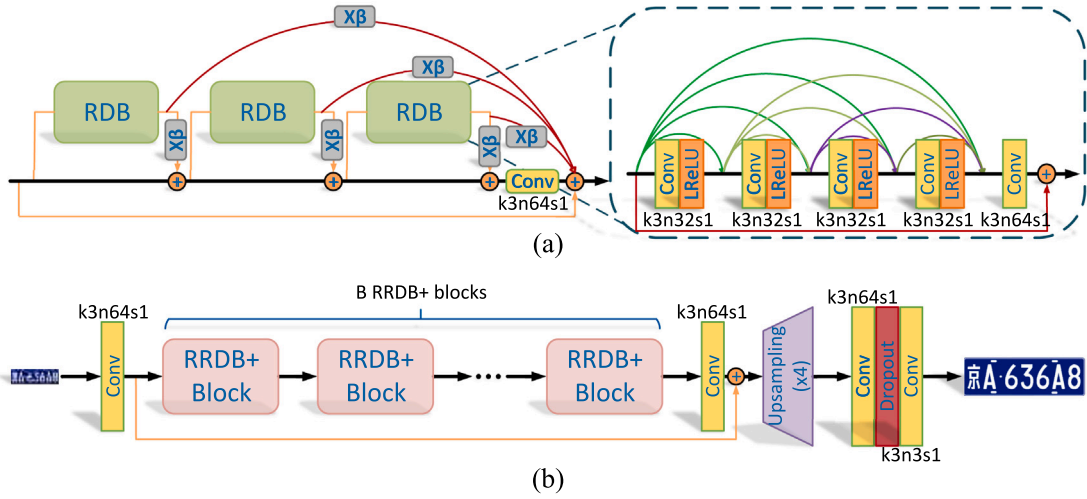


Fig. 8. Overall architecture of the generator: (a) RRDB+ Block; and (b) A dropout layer is added before the last convolutional layer of the generator. Where k, n and s respectively represent the kernel size, output feature map number and step size of each convolution layer.

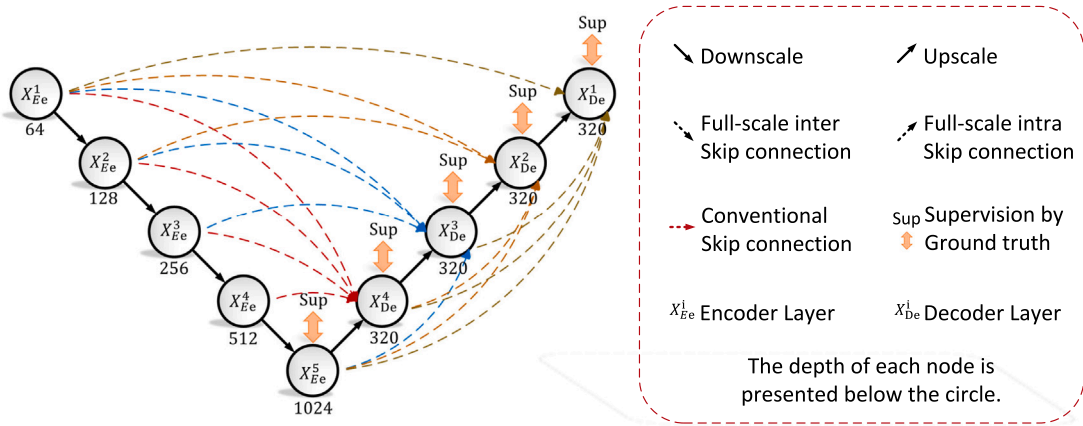


Fig. 9. Structure of U-Net3+ discriminator.

UNet and receives the feature map X_{Ee}^3 directly from the same scale encoder layer. Unlike UNet there is more than one skip connection. In particular, the top two jump connections pool down-sample the smaller scale encoder layers X_{Ee}^1 and X_{Ee}^2 through different maximum pooling operations to pass the lower-level semantic information of the underlying layer. The other two lower jump connections up-adopt X_{Ee}^4 and X_{Ee}^5 in the decoder by bilinear interpolation to amplify the resolution of the feature map.

3.3. Loss function

Since this paper focuses on the SR reconstruction of license plate images, the sharpness of the reconstructed characters is one of the critical metrics we must consider, which has a decisive effect on the downstream tasks. To improve the clarity of reconstructed image characters, this paper introduces a perceptual OCR loss based on the error in license plate recognition, which calculates the error between the OCR sequence of SR license plate images and the label sequence based on connectionist temporal classification (CTC), i.e.,

$$l_{OCR}^{SR} = -\log \left(\sum_{\pi \in \text{valid paths } (S,T)} p(\pi | S) \right), \quad (12)$$

where $S = O_{OCR}(I^{SR})$ is the OCR sequence of the SR license plate image, and T is the tag sequence. The OCR network uses the model provided by the open-source project HyperLPR3.

In addition, we cannot ignore the overall effect of the reconstructed images. For this reason, we retain the MAE loss in Real-ESRGAN [4], the perceptual loss based on VGG19, and the adversarial loss in this paper.

The L1 loss is given by

$$l_{MAE}^{SR} = \frac{1}{r^2 W H} \sum_{x=1}^{rW} \sum_{y=1}^{rH} |I_{x,y}^{HR} - G_{\theta_G}(I_{x,y}^{LR})|, \quad (13)$$

where W and H respectively denote the size of the input LR image, r denotes the scale factor, G_{θ_G} denotes the generator network parameterized by θ_G , I^{HR} and I^{LR} respectively denote the HR image as Ground truth and the LR image as degraded by I^{HR} , and x and y are respectively the horizontal and vertical coordinates of the image.

The perceptual loss based on VGG19 is

$$l_{VGG19/i,j}^{SR} = \frac{1}{W_{i,j} H_{i,j}} \sum_{x=1}^{W_{i,j}} \sum_{y=1}^{H_{i,j}} |I_{x,y}^{HR} - G_{\theta_G}(I_{x,y}^{LR})|, \quad (14)$$

where $W_{i,j}$ and $H_{i,j}$ describe the size of the feature map output from the j th convolutional layer in the i th VGG block of the VGG19 network. We use the output of the feature maps from the last convolutional layer of each VGG block in VGG19 to compute the perceptual loss as follows:

$$l_{percep}^{SR} = p_1 l_{VGG19/1,2}^{SR} + p_2 l_{VGG19/2,2}^{SR} + p_3 l_{VGG19/3,4}^{SR} + p_4 l_{VGG19/4,4}^{SR} + p_5 l_{VGG19/5,4}^{SR}, \quad (15)$$

where $p_i (i = 1, 2, \dots, 5)$ is the weight factor. The antagonistic loss is given by

$$l_{Gen}^{SR} = \sum_{n=1}^N -\log D_{\theta_D} (G_{\theta_G} (I^{LR})), \quad (16)$$

where D_{θ_D} denotes the discriminator network parameterized by θ_D .

3.4. Training strategy

We divided the training process into two stages. In the first stage, we trained the generator, RRDBNet+, targeting the highest PSNR using the loss function

$$l_{step1}^{SR} = \alpha l_{MAE}^{SR}, \quad (17)$$

In the second stage, we used RRDBNet+ as a pre-training model to reconstruct the character clarity of the image by combining perceptual OCR loss, L1 loss, VGG19-based perceptual loss, and adversarial loss for the generator and discriminator of LPSRGAN for joint training using the loss function

$$l_{step2}^{SR} = \alpha l_{MAE}^{SR} + \beta l_{percep}^{SR} + \gamma l_{Gen}^{SR} + \delta l_{OCR}^{SR}, \quad (18)$$

where α , β , γ and δ are the weights of each loss component.

4. Experiments

4.1. Datasets and implementation details

Training details. In this paper, we created a license plate image dataset, LicensePlateDataset10K, comprising 10,500 license plate images. The dataset includes genuine license plate images sourced from the CCPD2019 dataset [51] after affine transformation, alongside images synthesized by our team, with a ratio of 8:2. The training set from this dataset contains 10,000 images, the validation set contains 500 images and the HR patch for training is set to 128. We trained the model using 2X NVIDIA GeForce RTX 3090 with a total batch size 32, using the Adam optimizer [52]. We trained RRDBNet+ for 1000K iterations with a learning rate of 2×10^{-4} and LPSRGAN for 200K iterations with a learning rate of 1×10^{-4} . The first stage of LPSRGAN training has a loss function with weight $\alpha = 1.0$ and the second stage has a loss function with weights $\alpha = 0.01$, $\beta = 1.0$, $\gamma = 0.05$, $\delta = 0.05$ and weights of perceptual loss $\{p_1, p_2, p_3, p_4, p_5\} = 0.1, 0.1, 1.0, 1.0, 1.0$. Our generator has 23 identical RRDB+ blocks ($B = 23$), where the parameter of LeakyReLU activation is $\alpha = 0.2$, and the dropout probability of the dropout layer is $p = 0.5$. Our implementation is built upon the framework provided by BasicSR [53].

Degradation parameters. The present study investigates the relationship between the average license plate recognition rate and the number of stages in the degradation model. As shown in Fig. 10., degradation models with fewer than three stages have a negligible impact on the recognition rate, whereas models with more than three stages result in complete loss of character information in the license plate. In practice, we use a three-stage degradation model to simulate the degradation of license plate images to strike a balance between simplicity and effectiveness, with a magnification of 4 between LR and HR images. The probability of performing three stages of degradation is set to 1, 0.8, and 0.5, respectively. The parameter settings for each stage remain essentially the same. The probabilities of using the Gaussian kernel, generalized Gaussian kernel, and fuzzy plateau kernel for each stage are 0.7, 0.15, and 0.15, respectively. The size of the fuzzy kernel is chosen randomly from $\{7, 9, \dots, 21\}$. The fuzzy standard deviation σ is set as a random number in $[0.2, 3]$, and the shape parameters β of the generalized Gaussian kernel and the fuzzy plateau kernel are set as random numbers in $[0.5, 4]$ and $[1, 2]$, respectively. The ringing effect is synthesized at each stage with a probability of 0.1. The possibilities of upsampling, downsampling, and holding for each stage are 0.1, 0.8, and

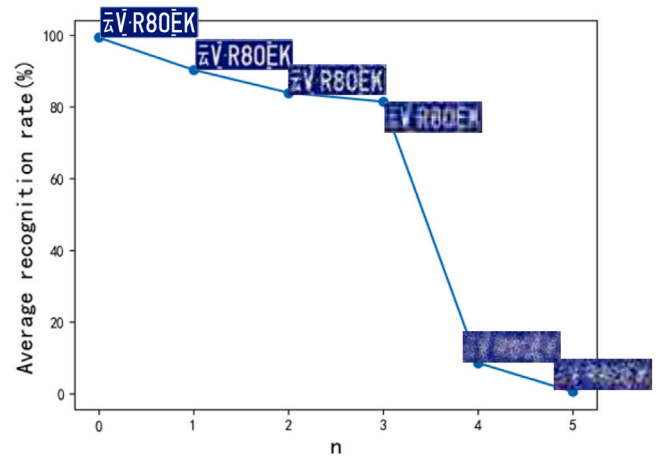


Fig. 10. Average recognition rate vs. number of stages n.

0.1, respectively, and the sampling range is $[0.25, 2]$. The probability of using Gaussian noise and Poisson noise in each stage is 0.5, and the standard deviation σ of Gaussian noise is set to a random number in $[1, 5]$. The scale of Poisson noise is set to a random number in $[2, 3]$. Greyscale noise has a probability setting of 0.8. Additionally, the JPEG compression quality factor is specified as a range of $[10, 30]$.

4.2. Experimental results and comparison

We compared our LPSRGAN with several state-of-the-art SR methods, including ESRGAN [1], ESRGAN+ [2], BSRGAN [3] and Real-ESRGAN [4], which we evaluated on a test set of LicensePlateDataset10K. In addition, considering license plate images as a specific type of scene text images, we extend the comparison to scene text super-resolution methods, including TSRN [5], RTSRN [6], and MARCONet [7]. Since the existing PSNR-oriented evaluation metrics and perceptual quality metrics do not reflect well the clarity of characters after license plate image reconstruction, we show several representative sample results using various methods in Fig. 11 for qualitative comparison.

Fig. 11 shows that our LPSRGAN significantly outperforms the previous methods in recovering character details. Specifically, each network achieves better reconstruction results for the first two simple samples (in the first two columns of the figure), but our LPSRGAN performs better in the details of Chinese characters. For the last four complex samples (in the last four columns), the images reconstructed by the other networks show significantly garbled images. But our LPSRGAN is still able to retain the detail information of the characters to the maximum extent, and although its results for the last two samples show some errors in character detail, they do not affect the overall recognition results.

In addition, we designed an objective evaluation metric based on license plate recognition without HR reference, which effectively evaluates the quality of reconstructed license plate images by calculating the average value of the difference between the correct character recognition rate of LR license plate images and SR reconstructed license plate images, i.e.,

$$\Delta R = \frac{1}{N} \sum_{i=1}^N [R(I_i^{SR}) - R(I_i^{LR})], \quad (19)$$

where R denotes the correct character recognition rate, and to ensure fairness we use HyperLPR3, an open-source high-performance license plate recognition framework from a third party, as the recognizer. I_i^{SR} denotes the i th SR reconstructed image, I_i^{LR} denotes the i th original LR image, and N is the total number of test images. We used this evaluation metric on the LicensePlateDataset10K dataset to compare



Fig. 11. Qualitative comparison of several representative samples.



Fig. 12. Effect of various SR methods on actual captured images.

with several state-of-the-art SR methods as reported in Table 1. Our LPSRGAN improves the recognition rate of license plate images from 81.42% to 93.90% with a ΔR of 12.48%, outperforming other image SR methods.

Although this evaluation metric can effectively evaluate the reconstruction results of license plate images or other character type images, it has some limitations. First, it cannot be used for evaluation of non-character images. Second, it requires that the character labels are known. In addition, this evaluation metric may also produce some

potential biases. For example, two different images with the same characters will be considered one image under this metric, and two images with similar characters may produce erroneous evaluation results.

In order to make up for the shortcomings of the evaluation metric proposed in this paper, as a supplement, we also used Peak Signal-to-Noise Ratio (PSNR) and Structural Similarity (SSIM) to evaluate the experimental results, as shown in Table 2. Due to some limitations of our model, which are described in detail in Section 4.5, our model does not achieve the best performance on PSNR and SSIM. It is of

Table 1

Comparison of the performance of various SR methods on the evaluation metric ΔR with the best performance denoted in bold.

Method	Average recognition rate (%)			
	LR	SR	ΔR	HR
ESRGAN [1]		82.21	0.79	
ESRGAN+ [2]		82.67	1.25	
BSRGAN [3]		84.30	2.88	
Real-ESRGAN [4]		89.00	7.58	
TSRN [5]	81.42	83.36	1.88	99.37
RTSRN [6]		85.28	3.86	
MARCONet [7]		88.91	7.49	
LPSRGAN(ours)		93.90	12.48	

concern that PSNR and SSIM may not finely capture human perceptual preferences due to their insensitivity to certain perceptible distortions, and an improvement in these metrics does not necessarily imply an improvement in the quality of the vision, so they are only used as a reference.

4.3. Generalization to real data

In addition to significantly improve the performance of license plate image SR, our proposed LPSRGAN has good generalization ability. Note that our model is trained entirely on synthetic HR-LR image pairs, but the LPSRGAN model trained on LicensePlateDataset10K performs well on actual captured LR license plate images, as shown in Fig. 12.

In this case, the capture device is MI 11 PRO. Fig. 12 shows the captured complete image on the left, and the LR license plate image after perspective transformation and the SR image after application of various models on the right. As can be seen from the figure, ESRGAN [1] and ESRGAN+ [2] tend to generate overly smooth results, BSRGAN [3] introduces more interfering information, and Real-ESRGAN [4] is unable to effectively reconstruct character information in some cases, especially Chinese characters. As scene text super-resolution methods, TSRN [5], RTSRN [6], and MARCONET [7] exhibit strong performance in enhancing character details, yet they tend to overlook the overall visual coherence of the image. Compared with other methods, our LPSRGAN effectively recovers most of the character information in the actual captured LR license plate image. Although certain characters show some deviations, they have little effect on the recognition results.

4.4. Ablation experiment

We conducted ablation experiments to verify the effectiveness of each module proposed in this paper. As reported in Table 3, setting (a) uses Real-ESRGAN [4] as a benchmark, and the symbol $\checkmark$ indicates that the corresponding module is applied.

To demonstrate the effect of each proposed module, we incorporated them individually and in combination with a benchmark. Firstly, the integration of n-RCD, as illustrated in setting (b), led to improvements of 0.61, 2.01, and 0.1089 in the ΔR , PSNR, and SSIM metrics, respectively. This suggests that n-RCD is capable of covering a broader degradation space and effectively enhancing the generalization ability of the model. Secondly, the inclusion of RRDBNet+, as depicted in setting (c), resulted in a 1.21 increase in the ΔR metric, but it concurrently decreased performance on PSNR and SSIM, indicating that RRDBNet+ can fully leverage information from intermediate features, albeit introducing some interference. Finally, the addition of the perceptual OCR loss, as shown in setting (e), led to respective performance gains of 0.83, 0.85, and 0.0419 in the three metrics, signifying its effectiveness in enhancing the clarity of reconstructed image characters. Furthermore, we conducted a study on the combination of these modules, revealing that their integration further enhanced the performance of the model, as evident in setting (d), (f) and (g). The proposed method

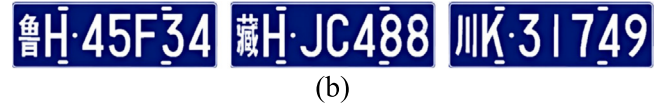


Fig. 13. Effectiveness of n-RCD: (a) using RRDB+ without n-RCD; and (b) using RRDB+ with n-RCD.

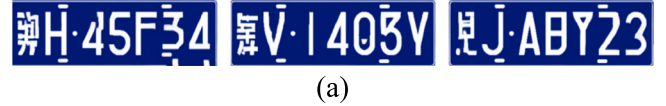


Fig. 14. Validity study on RRDBNet+ generator: (a) LPSRGAN without RRDB+; and (b) LPSRGAN with RRDB+.

with the default setting (h) obtains the best performance of 12.48 on ΔR metric.

In addition, we also provide visual results of ablation experiments in the next three subsections to more intuitively compare the effects of each module proposed in this paper.

4.4.1. n-RCD

To ensure controllability, we performed an ablation study of the degradation model on RRDBNet+ because it is more stable and better reflects the impact of degradation. To generate training pairs, we replaced n-RCD with the second-order degradation model in Real-ESRGAN [4]. As shown in Fig. 13, the model trained with the second-order degradation model cannot effectively recover the character details of the license plate image, especially the Chinese characters, while RRDBNet+ can handle this case.

4.4.2. RRDBNet+

To investigate the effectiveness of RRDBNet+, we first used RRDBNet as a generator to train LPSRGAN. As shown in Fig. 14, since RRDBNet does not consider the output features of the intermediate module, its reconstructed images have limited character integrity. However, the RDB feature aggregation module in RRDBNet+ effectively handles this situation.

4.4.3. Perceptual OCR loss

We investigated the role of perceptual OCR loss by first removing this in the loss function. As reported in Fig. 15, due to the lack of text constraints, the model without the perceptual OCR loss forms garbled characters in the reconstructed images, especially the Chinese text. Adding this loss effectively generates the correct characters.

4.5. Limitation

While LPSRGAN demonstrates the ability to recover the majority of LR license plate images, it still exhibits certain limitations, as depicted in Fig. 16. First, due to the use of the RDB feature aggregation module in the generator, it may introduce some trivial interference to the reconstructed images. Second, due to the inclusion of perceptual OCR loss, it may generate incorrect text in complex degradation cases. These drawbacks significantly impact the practical application of LPSRGAN and require urgent attention in future work.

Table 2

Comparison of the performance of various SR methods using PSNR and SSIM, with the best and second-best results bolded and underlined, respectively.

Metric	Method								HR
	ESRGAN [1]	ESRGAN+ [2]	BSRGAN [3]	Real-ESRGAN [4]	TSRN [5]	RTSRN [6]	MARCONet [7]	LPSRGAN(ours)	
PSNR	13.47	15.74	15.19	18.28	11.83	11.21	13.34	<u>18.14</u>	∞
SSIM	0.4504	0.6482	0.6888	0.8098	0.3924	0.4236	0.4423	<u>0.7994</u>	1

Table 3

Ablation analysis in terms of ΔR , PSNR and SSIM, with the best and second best results are bolded and underlined, respectively.

Setting	n-RCD	RRDBNet+	Perceptual OCR loss	ΔR	PSNR	SSIM
(a)				7.58	18.28	0.8098
(b)	✓			8.19	<u>20.29</u>	<u>0.9187</u>
(c)		✓		8.79	15.93	0.6302
(d)	✓	✓		9.96	17.16	0.6695
(e)			✓	8.41	19.13	0.8517
(f)	✓		✓	9.61	20.57	0.9276
(g)		✓	✓	<u>10.27</u>	16.84	0.7831
(h) default	✓	✓	✓	12.48	18.14	0.7994

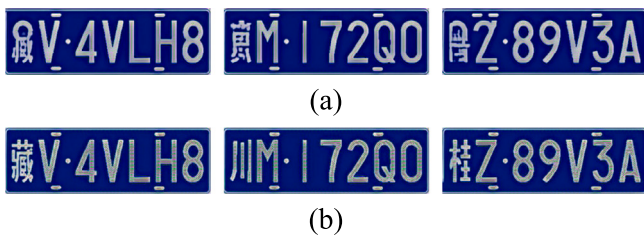


Fig. 15. Effectiveness of perceptual OCR: (a) LPSRGAN without perceptual OCR loss; and (b) LPSRGAN with perceptual OCR loss.



Fig. 16. Limitations of LPSRGAN: (a) Interference information; and (b) Generated incorrect text.

5. Conclusion

This paper presents a SR algorithm for license plate images based on GANs, which effectively improves the recognition rate of LR license plate images by introducing a new image degradation model and LPSRGAN combined with perceptual OCR loss for license plate images.

The experimental results demonstrate that the algorithm improves the recognition rate of the SR license plate images by 12.48%, and the recognition rate of the reconstructed images reaches 93.90% compared to the original LR license plate images. The performance of the algorithm on captured data indicates its strong practicality and promising application prospects.

CRediT authorship contribution statement

Yuecheng Pan: Data curation, Formal analysis, Investigation, Methodology, Software, Validation, Visualization, Writing – original draft, Writing – review & editing. **Jin Tang:** Conceptualization, Funding acquisition, Methodology, Project administration, Resources, Supervision. **Tardi Tjahjadi:** Writing – original draft, Writing – review & editing.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

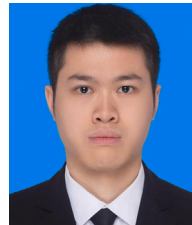
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